SDPay

Mục lục

1. Overview

1.1 Purpose

Documentation provided to partners to connect to the Payment system. This document is only for technical staff, programmers responsible for system design, solution integration into Website, Mobile App, or e-commerce application

1.2 Scope

Documentation provided to partners to connect to the Payment system. This document is only for technical staff, programmers responsible for system design, solution integration into Website, Mobile App, or e-commerce application

2. Platform

2.1 Recharge platform

Step 1: Get transfer information

- User enters to get transfer information (Importan It is the content that User needs to enter when making a transfer, in the content contains username to distinguish which user performs the transfer)
- The Partner takes this information and displays it on the screen for the customer to transfer

Step 2: User makes the transfer according to the transfer

- Partner receives customer's electronic order and sends Request to SDPay, SDPay Response for transfer information

Step 3: Announcement of transaction results

- After receiving the order, SDPay will put it in the queue to process the User's transfer

Step 4: Transaction settlemen

- SDPay will notify the Partner of the transaction results via the Callback link

2.2 Cash

Step 1 Enter order information cash out

- User accesses the Partner's website and App to place a withdrawal order
- The user enters the account information, the bank to withdraw money, the amount to withdraw

Step 2: The SHOP system places cash out orders and announces the results

- SDPay system places cash out orders and announces the results

Step 3: Transaction settlement

- Total payment data will be confirmed by both parties after each reconciliation cycle

3. Recharge API (WebHook - JSon-Post)

3.1 Get List Bank Code

- **Desription: get list of banks. can skip this step if you use Bankcode=random in api order**
- **URL: https://bankgate.sdpay.info/bankin/info.ashx**
- **Method:** POST
- **Content-Type**: application/json
- **InPut:** Json Object

Object RequestData:

```
public class RequestData
 {
       public string PartnerCode { get; set; }

 }
```

Paramters	Data Type	Description
PartnerCode	string	PartnerCode (**will provide when connected)**

JSON OutPut:


```
{
      "ResponseCode": 1,
      "Description": "Transaction is successful",
      "ResponseContent": "[{\"BankCode\":\"ACB\",\"Name\"\"Ngân hang Á Châu
      ACB\"},{\"BankCode\":\"VCB\",\"Name\":\" Vietcombank \"}]",

}
```


Described as follows:

```
public class APIResponse
    {
        public int ResponseCode { get; set; }
        public string Description { get; set; }
        public string ResponseContent { get; set; }

    }
```

Paramters	Data Type	Desciption
ResponseCode	string	Status of transaction (1 sucess)
Description	string	Description
ResponseContent	string	Content of the transaction result (In case the result fails,the content of this string will not be available)

ResponseContent: JSON of List Object

```
public class BankAccount
    {
        public string BankCode { get; set; }
        public string Name { get; set; }

    }
```

Paramters	Data Type	Description
BankCode	string	Code of bank
Name	string	Name of bank

3.2 Create Order

- **Description: Create order show receive bank**

- **URL: https://bankgate.sdpay.info/bankin/order.ashx**
- **Method:** POST
- **Content-Type**: application/json
- **In Put:** Json Object

Object RequestData:

```
public class RequestData
 {
       public string PartnerCode { get; set; }
       public int Amount { get; set; }
       public string BankCode { get; set;
       public string CallbackUrl { get; set; }
       public string RefCode { get; set; }
```

```
        public string Signature { get; set; }
    }
```

Paramters	Data Type	Description
PartnerCode	string	PartnerCode (will provide when connected)
BankCode	string	Bank code that the User chooses (get the list in api section 3.1) **In case the partner does not want to choose a bank.** **You can transmit BankCode=random. The system will randomly return information to a receiving account**
RefCode	string	Reference Code (Usually will be the Transaction Id of the counterparty, this should be unique)
Amount	int	Amount of tranfer
CallbackUrl	String	Callback url
Signature	string	Signature: MD5(**PartnerCode** + **BankCode** + **Amount** + **RefCode** + **CallbackUrl** + **partnerKey**) partnerKey : will provide when connected

- **Out Put:**


```
{
    "ResponseCode": 1,
    "Description": "Transaction is successful",
    "ResponseContent":
    "{\"Status\":\"1\",\"BankCode\":\"ACB\",\"Url\":\"xxx\",\"QRCode\
    ":\"xxxx\",\"AccountNumber\":\"123123123\",\"AccountName\":\"Nguy
```

```
    en Van
    A\",\"Amount\":10000,\"RefCode\":\"1223\",\"OrderNo\":\"AA123123\
    "}"

}
```

Described as follows:

```
public class APIResponse
    {
            public int ResponseCode { get; set; }
            public string Description { get; set; }
            public string ResponseContent { get; set; }

    }
```

Paramters	Data Type	Description
ResponseCode	string	Status of transaction (1 sucess)
Description	string	Description
ResponseContent	string	Content of the transaction result (In case the result fails,the content of this string will not be available)

ResponseContent: JSON of List Object

```
    public class Order
    {

            public string BankCode { get; set; }
            public string AccountNumber { get; set; }
            public string AccountName { get; set; }
            public int Amount { get; set; }
            public int RefCode { get; set; }
            public string OrderNo { get; set; }
            public string Url { get; set; }
            public string QRCode{ get; set; }
    }
```

Thuộc tính	Data Type	Description
BankCode	string	Code of bank.
AccountNumber	string	Account number to receive
AccountName	string	Account Name to receive

Amount	Int	Amount of Tranfer
Url	string	payment page address
QRCode	String	QR Code
RefCode	string	The partner's reference code is transmitted in the Request order section
OrderNo	string	**The transaction code generated by SDPay is also the transfer content that the User needs to enter when transferring money**

*Note

- To ensure stable transaction processing and minimize timeouts caused by network latency, please configure the timeout duration for each API request to 60 seconds.

Exam

```
PHPcurl_setopt($ch, CURLOPT_CONNECTTIMEOUT, 60);
url_setopt($ch, CURLOPT_TIMEOUT, 60);
```

```
 C#
var request = (HttpWebRequest)WebRequest.Create(url);
request.Timeout = 60000;
```

3.3 Callback (Webhook build from partner)

- **Description:** This is a partner API built according to the below standards to receive
- **URL:** CallbackUrl parameter at api 3.2
- **Method:** POST
- **Content-Type**: application/json
- **In Put:**

JSON InPut:

```json
{
    "ResponseCode": 1,
    "Description": "Transaction is successful",
    "Amount": 1000000,
    "RefCode": "xxxx",
    "OrderNo": "xxxx",
    "OrderInfo": "xxxx",
    "Signature": "778a2c01dbdac3c4310deb1987c9e6e2"
}
```

Body JSON:

```
public class APIResponse
        {
                public int ResponseCode { get; set; }
                public string Description { get; set; }
                public string RefCode { get; set; }
                public string OrderNo { get; set; }
                public int Amount { get; set; }
                public int Fee { get; set; }
                public string OrderInfo { get; set; }
```

Paramters	Data Type	Description
ResponseCode	string	Status of Result
Description	string	Description
RefCode	string	Reference Code (Usually will be the Transaction Id of the counterparty, this should be unique)
OrderNo	string	The name of the user performing the money transfer on the system
Amount	int	Amount actually received
Fee	int	Amount Fee
OrderInfo	String	Transaction of bank
Signature	string	Signature: MD5(**ResponseCode** + **Description** + **RefCode** + **Amount** + **partnerKey**) partnerKey : will provide when connected

- **OutPut:** After receiving the Callback, the partner processes the transaction and returns the string delimiter code|description content as follows
  - **success:** 1|success
  - **failed:** -1|failed

3.4 GetTransction

- **Description:** get tranfer status
- **URL:** https://bankgate.sdpay.info/bankin/getdetail.ashx
- **Method:** Post
- **In Put:** Json Object

Object RequestData:

```
public class RequestData
 {
        public string PartnerCode { get; set; }
        public string RefCode { get; set; }
        public string Signature { get; set; }



 }
```

Paramters	Data Type	Description
PartnerCode	string	PartnerCode (will provide when connected)
RefCode	string	Reference Code
Signature	string	MD5(**partnerCode** + **RefCode**+ **partnerKey**) partnerKey : will provide when connected

- **Out Put:**

```
{
   "ResponseCode": 1,
   "Description": "Transaction is successful",
   "ResponseContent":
   "{\"Amount\":20000,\"RefCode\":\"2\",\"TransactionID\":\"1000011\
   ",\"LasTime\":\"\/Date(1752160017623)\/\"}"
}
```

```
public class APIResponse
   {
         public int ResponseCode { get; set; }
         public string Description { get; set; }
```

```
        public string ResponseContent { get; set; }
}
```

Paramters	Data Type	Description
ResponseCode	string	Status of transaction. 1 is success 0: is process - 320 transaction not exist -357 bank info wrong -1 transaction is fail
Description	string	Description
ResponseContent	string	Contents of transaction results when ResponseCode =1

ResponseContent: JSON of Object when ResponseCode =1

```
public class TransactionInfo
{
        public string RefCode { get; set; }

        public int Amount { get; set; }
        public dateime LastTime { get; set}

}
```

Paramter	Data Type	Description
RefCode	string	Reference Code (Usually will be the Transaction Id of the counterparty, this should be unique)
Amount	int	Amount actually received
LastTime	datetime	Time is transaction success

▪

3.5 Table ResponseCode

Only ResponeCode = 1 is the case of success, the rest of the codes will be specific failures described in the following error code table

Code	Description
1	Transaction Successful

-1	Transaction Failed
-323	Parameter Invalid
-313	Signature Invalid
-317	Service Not Exists
-312	Partner Not Exists Not Active
-301	System Error
-310	Access Denied
-326	Transaction Timeout
-357	Bank Account is Wrong
-356	Bank Amount Invalid
-300	System Maintain

4. Cash API (WebHook - JSon-Post)

4.1 GetBalance

Description get balance

- **URL: https://bankgate.sdpay.info/ServiceHandler/GetBalanceV2.ashx**
- **Method: HTTP Get**
- **In Put 2 param**

Paramters	Data Type	Mô tả
partnerCode	string	PartnerCode (will provide when connected)
type	String	type =json
signature	string	MD5(**partnerCode +panerKey**)

Out Put: balance json

```
{"Balance":10000}
```

```
public class BalanceResponse
{
        public long Balance { get; set; }
```

```
}
```

Paramters	Data Type	Description
Balance	long	Balance of partner

4.2 Cash

- **Description:** create order to cash out
- **URL: https://bankgate.sdpay.info/bankout/cash.ashx**
- **Method:** POST
- **In Put:** Json Object

Object RequestData:

```
public class RequestData
 {
        public string PartnerCode { get; set; }
        public string AccountName { get; set; }
        public string AccountNumber { get; set; }
        public string BankCode { get; set; }
        public string RefCode { get; set; }
        public int Amount { get; set; }
        public string CallbackUrl { get; set; }
        public string Signature { get; set; }



 }
```

Paramters	Data Type	Description
PartnerCode	string	PartnerCode (will provide when connected)
BankCode	string	Bank code: in the table below for bank BankCode=MOMO when using MOMO wallet
AccountName	string	Account Nanme receiving money
AccountNumber	string	Account number receiving money
CallbackUrl	Strign	Callback Url
RefCode	string	Reference Code (Usually will be the Transaction Id of the counterparty, this should be unique)

Amount	int	Amount received by the user
Signature	string	MD5(**partnerCode** + **AccountNumber** + **AccountName** + **BankCode** + **Amount** + **RefCode**+ **CallbackUrl** + **partnerKey**) partnerKey : will provide when connected

- **Out Put:**

```
{
   "ResponseCode": 1,
   "Description": "Transaction is successful",
   "ResponseContent": ""
}
```

```
public class APIResponse
   {
          public int ResponseCode { get; set; }
          public string Description { get; set; }
          public string ResponseContent { get; set; }

   }
```

Paramters	Data Type	Description
ResponseCode	string	Status of transaction results. 1 means successfully receiving the order
Description	string	Description

*Note

- To ensure stable transaction processing and minimize timeouts caused by network latency, please configure the timeout duration for each API request to 60 seconds.

- If you receive an HTTP 502, 504, 520, or 524 error, the order may have already been successfully received by our system=> Please check the order status or contact us before resubmitting the request.

4.3 Callback

- **Description:** This is a partner API built according to the below standards to receive
- **URL:** it is `CallbackUrl` param at 4.1
- **Method:** POST
- **Content-Type**: application/json
- **In Put:**

JSON InPut:


```
{
    "RefCode":"xxxx",
    "TransactionID":"xxx",
    "Amount":10000,
    "OrderInfo":"xxx-2542","
    "ResponseCode":1,"
    Description":"Transaction is successful","
    Signature":"96f7843da57ea940e3ef1a38b08e164b"
}
```


Body of json:

```
public class APIResponse
        {
                public int ResponseCode { get; set; }
                public string Description { get; set; }
                public string RefCode { get; set; }
                public string TransactionID{ get; set; }
                public int Amount { get; set; }
                public string Signature { get; set; }
        }
```

Paramters	Data Type	Description
ResponseCode	string	ResponseCode =1 is success ResponseCode < 0 is fail
Description	string	Description
TransactionID	string	Transaction of system
RefCode	string	Reference Code (Usually will be the Transaction Id of the counterparty, this should be unique)
Amount	int	Amount actually received
Signature	string	MD5(**ResponseCode**+ **Description**+ **RefCode** + **partnerKey**)

- **OutPut:** After receiving the Callback, the partner processes the transaction and returns the string delimiter code|description content as follows
  - **success:** 1|success
  - **failed:** -1|failed

4.4 GetTransction

- **Description:** get tranfer status
- **URL:** https://bankgate.sdpay.info/bankout/getdetail.ashx
- **Method:** Post
- **In Put:** Json Object

Object RequestData:

```
public class RequestData
{
    public string PartnerCode { get; set; }
    public string RefCode { get; set; }
    public string Signature { get; set; }


}
```

Paramters	Data Type	Description
PartnerCode	string	PartnerCode (will provide when connected)
RefCode	string	Reference Code
Signature	string	MD5(**partnerCode** + **RefCode**+ **partnerKey**) partnerKey : will provide when connected

- **Out Put:**

```
{
    "ResponseCode": 1,
```

```
    "Description": "Transaction is successful",
    "ResponseContent":
    "{\"Amount\":20000,\"RefCode\":\"2\",\"TransactionID\":\"1000011\
    ",\"LasTime\":\"\/Date(1752160017623)\/\"}"
}
```

```
public class APIResponse
   {
          public int ResponseCode { get; set; }
          public string Description { get; set; }
          public string ResponseContent { get; set; }

}
```

Paramters	Data Type	Description
ResponseCode	string	Status of transaction. 1 is success 0: is process - 320 transaction not exist -357 bank info invalite -1 transaction is fail
Description	string	Description
ResponseContent	string	Contents of transaction results when ResponseCode =1

ResponseContent: JSON of Object when ResponseCode =1

```
public class TransactionInfo
{
       public string RefCode { get; set; }

       public int Amount { get; set; }
       public dateime LastTime { get; set}

}
```

Paramter	Data Type	Description
RefCode	string	Reference Code (Usually will be the Transaction Id of the counterparty, this should be unique)
Amount	int	Amount actually received

LastTime	datetime	Time is transaction success

4.5 Table ResponseCode

Only ResponeCode = 1 is the case of success, the rest of the codes will be specific failures described in the following error code table.

Code	Description
1	Transaction Successful
-1	Transaction Failed
-323	Parameter Invalid
-313	Signature Invalid
-317	Service Not Exists
-312	Partner Not Exists Not Active
-301	System Error
-310	Access Denied
-326	Transaction Timeout
-357	Bank Account Invalid
-5	Transaction Cancel
-356	Bank Amount Invalid
-350	Bank Info Invalid
-352	Bank Code Invalid
-358	Bank Code Maintain
-300	System Maintain
-51	Balance Not Enough

4.6 Table Bank Code

ICB - VietinBank

VCB - Vietcombank

BIDV - BIDV

VBA - Agribank

OCB - OCB

MB - MBBank

TCB - Techcombank

ACB - ACB

VPB - VPBank

TPB - TPBank

STB - Sacombank

HDB - HDBank

VCCB - VietCapitalBank

SCB - SCB

VIB - VIB

SHB - SHB

EIB - Eximbank

MSB - MSB

SGICB - SaigonBank

BAB - BacABank

PVCB - PVcomBank

Oceanbank - Oceanbank

NCB - NCB

SHBVN - ShinhanBank

ABB - ABBANK

VAB - VietABank

NAB - NamABank

PGB - PGBank

VIETBANK - VietBank

BVB - BaoVietBank

SEAB - SeABank

COOPBANK - COOPBANK

LPB - LienVietPostBank

KLB - KienLongBank

KBank - KBank

UOB - UnitedOverseas

SCVN - StandardChartered

PBVN - PublicBank

NHB HN - Nonghyup

IVB - IndovinaBank

IBK - HCM - IBKHCM

IBK - HN - IBKHN

VRB - VRB

WVN - Woori

KBHN - KookminHN

KBHCM - KookminHCM

HSBC - HSBC

HLBVN - HongLeong

GPB - GPBank

DOB - DongABank

DBS - DBSBank

CIMB - CIMB

CBB - CBBank

CITIBANK - Citibank

KEBHANAHCM - KEBHanaHCM

KEBHANAHN - KEBHANAHN

MAFC - MAFC

VBSP - VBSP